

CS 449 – Lecture 2 Note Template

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1 tar archives

- `tar` is used to _____
- Why will this course use `tar`? _____
- Create an archive: `tar -czf` _____
- Extract an archive: `tar -xzf` _____
- What do the flags `c`, `x`, `z`, and `f` mean? _____

2 C syntax parsing

- C is parsed _____
- “Before” means _____
- In Java, what still must be declared before use? _____
- In Java, what does *not* need to be declared above its use? _____
- In C, list three things that must appear above their use: _____

```
/* Why would this fail to compile in C? */
/* ----- */

int main(void) {
    foo();
}

void foo(void) {
    /* ----- */
}
```

3 Getting input for a C program

3.1 Command-line arguments

- 1st mechanism: _____
- `argv` is _____
- Type of `argv`: `char * argv[]` What is each element? _____
- `argv[0]` is always _____
- Example program from slides: _____

```
int main(int argc, char *argv[]) {  
    // argc is _____  
    // argv[1] is _____  
}
```

3.2 Console input (`scanf`)

- 2nd mechanism: _____
- Example program: _____
- `scanf` waits for _____
- The format string tells `scanf` _____
- Why do you pass `&variable`? _____
- What does a return value of 0 mean? _____

```
int n;  
int matched = scanf("%d", /* what goes here? */);  
/* matched == _____ */
```

4 Streams

- 3rd mechanism: _____
- FIFO means _____
- Name the three console streams and what each is for:
 - `stdin`: _____
 - `stdout`: _____

– stderr: _____

- Other stream examples from lecture: _____

Diagram: program, OS buffer, and a console stream

Sketch how data moves from your program through a buffer to stdin/stdout/stderr. Mark where a flush happens.

4.1 Buffering and explicit streams

- Streams are buffered by _____
- You must _____ to actually transfer data.
- Implicit functions such as `printf` use _____
- `fprintf` lets you _____

```
int printf(const char *format, ...);
int fprintf(FILE *stream, const char *format, ...);
/* When would you use fprintf instead of printf? */
/* _____ */
```

4.2 Shell redirection

- In Unix, files and streams are _____
- `>` does _____ Example: `ls > files.out`
- `>>` does _____ Example: `ls >> all_files.out`
- `<` does _____ Example: `wc < files.out`
- `|` does _____ Example: `ls | grep .c | wc`

5 Special streams

- The OS exposes special features as _____
- `/dev/null`: reads _____; writes _____
- `/dev/zero`: reads _____
- How is `/dev/zero` different from `/dev/null`? _____
- `/dev/random`: reads _____

6 File I/O in programs

- Command line and console streams are handled by _____
- Files you open yourself must be _____
- `fopen` returns _____ on error.
- After a text file is open, you can treat it like _____
- Text-file cousins of `printf/scanf`: _____

```
FILE *file = fopen(name, /* mode? */);
if (file == NULL) {
    /* ----- */
}
/* use fprintf / fscanf here */
fclose(file);
```

7 ASCII files vs raw values

- ASCII maps _____ to _____
- The character "7" is _____
- The integer 7 in memory is _____

Diagram: memory bytes vs file bytes for `fprintf("%d\n", i)`

For $i = 7$ and $i = -7$, write the memory representation and the bytes that land in the file. Slides used $0x00000007$ vs $0x37$, $0x0a$ and $0xffffffff9$ vs $0x2d$, $0x37$, $0x0a$.

8 Explicit file I/O

- An application can only operate on data that is _____
- The programmer must copy data _____
- Files are like arrays of _____
- File access is **explicit**. That means _____
- Memory access is often **implicit**. That means _____

8.1 fread and fwrite

- fread copies _____
- fwrite copies _____
- Arguments to think through: `ptr`, `size`, `nitems`, `stream`

```
int x = 5;
FILE *outfile = fopen(argv[1], "wb");
fwrite(&x, sizeof(x), 1, outfile);
/* memory contains ----- */
/* file contains ----- */
fclose(outfile);
```

9 File offsets

- Files are arrays of bytes, but `fread/fwrite` do not take _____
- The current file position is _____
- After you read 5 bytes, the position _____
- `fseek(fstream, 16, SEEK_SET)` moves _____

Diagram: file as a byte array with a moving position

Draw a file of bytes. Mark the current position, then show where it lands after reading 5 bytes and after `fseek(..., 16, SEEK_SET)`.

This template may not cover every topic from the source slides.